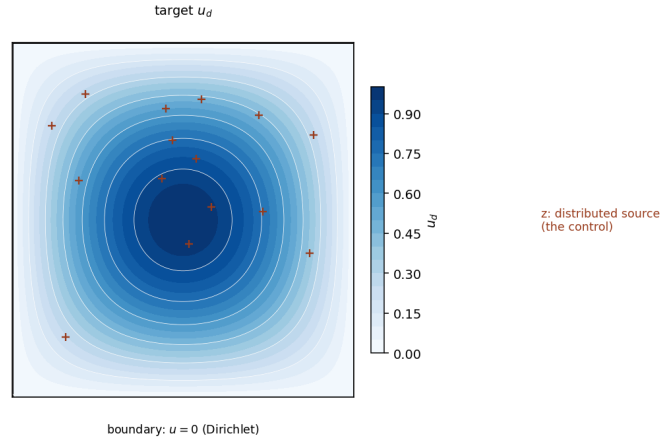


Assignment: Linear-Quadratic Optimal Control of the Poisson Equation

PDE-Constrained Optimization — Finite Element Method

1 Problem description

Consider the Poisson equation $-\Delta u = z$ on the unit square $\Omega = (0,1)^2$, with homogeneous Dirichlet boundary conditions. Instead of z being a given source term, here z is a **control**: it is chosen, not prescribed, so that the resulting state u is as close as possible to a given target field u_d , while keeping z itself small (large sources are penalized).



This is a PDE-constrained optimization problem: minimize a cost functional over z , subject to u solving the PDE for that z . The two ingredients are coupled — changing z changes u , which changes the cost — and the coupling is what makes the problem nontrivial to solve.

2 Governing equations

Optimization problem. Find z minimizing

$$J(u, z) = \frac{1}{2} \int_{\Omega} (u - u_d)^2 d\Omega + \frac{\alpha}{2} \int_{\Omega} z^2 d\Omega \quad \text{subject to} \quad -\Delta u = z \text{ in } \Omega, \quad u = 0 \text{ on } \partial\Omega$$

First-order optimality system. A minimizer (u, z) satisfies, together with an adjoint variable p :

$$\begin{cases} -\Delta u = z & \text{in } \Omega, \quad u = 0 \text{ on } \partial\Omega & \text{(state equation)} \\ -\Delta p = u - u_d & \text{in } \Omega, \quad p = 0 \text{ on } \partial\Omega & \text{(adjoint equation)} \\ \alpha z + p = 0 & \text{in } \Omega & \text{(optimality / stationarity condition)} \end{cases}$$

Weak forms. With $A(\cdot, \cdot)$ the usual Dirichlet bilinear form $A(w, v) = \int_{\Omega} \nabla w \cdot \nabla v d\Omega$:

$$A(u, v) = \int_{\Omega} z v d\Omega \quad \forall v, \quad A(p, v) = \int_{\Omega} (u - u_d) v d\Omega \quad \forall v$$

3 Task

1. Build a structured triangular mesh of $\Omega = (0, 1)^2$, and assemble the stiffness matrix A (P1, standard Dirichlet bilinear form) and the mass matrix M . Use the same A for both the state and adjoint equations, since $-\Delta$ with homogeneous Dirichlet boundary conditions is self-adjoint. M is needed for the source terms $\int z v d\Omega = (Mz)_i$ and $\int (u - u_d)v d\Omega = (M(u - u_d))_i$.
2. Implement a gradient descent loop on z : at each iteration, solve the state equation for u , solve the adjoint equation for p , form the reduced gradient $g = \alpha z + p$, update $z \leftarrow z - \tau g$.
3. **Gradient check.** At a random z , verify that g matches a finite-difference approximation of $\nabla_z J$.
4. **Recoverability test.** Pick any nonzero $z_{\text{true}}(x, y)$, solve the state equation for it to obtain u_d (so that u_d is exactly realizable by construction). Solve the optimal control problem for each α in Section 4 and verify that $z_\alpha \rightarrow z_{\text{true}}$ as $\alpha \rightarrow 0$.

4 Data

Symbol	Value	Meaning
Ω	$(0, 1)^2$	domain (unit square)
$u_d(x, y)$	$\sin(\pi x) \sin(\pi y)$	target state
α	$10^{-1}, 10^{-2}, 10^{-3}$	regularization weight (try all three)
τ	≈ 0.1	gradient descent step size (halve if it diverges)